

Instructions

Time Limit: 80 minutes of work time and 15 minutes after pencils/pens down to organize pages and put them in the envelope.

Maximum Score: 205

ATTENTION: This packet only contains problems. Do not submit anything from this packet!

Instructions: Problems that use the words “compute”, “list”, “identify”, “write down”, or “draw” only call for an answer; no explanation or proof is needed. Problems that use the word “find” do not require proof unless otherwise stated. All other questions require explanation or proof unless otherwise stated.

- Each team will receive exactly one packet with answer boxes (**NOT** this packet!). The boxes on the answer packet are all appropriately sized for the length of the question’s answer. Each team will also be provided with unlimited scratch paper.
- Write all answers in the corresponding provided box (the box that comes directly after the question). Do **NOT** write on the back of any pages of the answer packet. Any work written on the back of packet pages will not be graded.
- If a single problem has multiple parts, all answers must go into the same specified box. Problems with multiple boxes will be clearly specified. You may not write answers to more than one question in a box or use a different question’s box to continue your answer.
- Use blank scratch paper to work out problems. You may grab more scratch paper as needed. Afterward, it is recommended to transfer your final answers into the boxes in dark pencil. Your team will not get another packet with boxes if someone used a pen.
- If your team needs more room for a problem, indicate this in the corresponding answer box. Then, use one piece of your provided scratch paper to continue the answer. Label the page with your team number, the problem number, and the page number of the current problem (i.e. if you submit two pages of scratch paper as extra pages for a problem, label them as $[1/2]$ and $[2/2]$). **DO NOT PUT MULTIPLE QUESTIONS ON THE SAME PAGE.** Extra pages with multiple questions on the same page will not be graded.
- You may only use **one** side of each extra page. If an extra page has work on two sides, neither side will be graded.
- Each team will submit **all 29 pages** in the answer packet even if nothing has been written. They **must** be in the original page order. Misordered pages will not get graded and your score for those problems default to 0. Absolutely no other pages are to be submitted!

In your solution for a given problem, you may cite the statements of earlier problems (but not later ones) without additional justification, even if you haven’t solved them. The problems are ordered by content, **NOT DIFFICULTY**. It is to your advantage to attempt problems from throughout the test. While completing the round, you should not consult the internet or any materials outside of the content of this test (including results not covered in this power round). You may not use calculators.

Preliminaries

Important. In this power round, we will assume all sets are finite unless otherwise stated.

In this power round, we will be concerned with *matroids*, which can be thought of as a generalization of graphs. Before we make any definitions, however, we need to establish some notation and conventions which will be important for the rest of the round.

- A *set* is a collection of objects that does not contain duplicates. For instance, $\{6, 7\}$ is a set, while $\{6, 6, 7\}$ isn't.
- We use $:=$ as a definitional equality. That is, if we want to define S as the set $\{1, 2\}$, we would write $S := \{1, 2\}$.
- The *empty set* is the set containing nothing. It is denoted as $\emptyset$ and is equivalent to $\{\}$.
- If x is contained in a set S , we will write $x \in S$.
- We will use set-builder notation. That is, if we want to describe the set S of elements satisfying property $\mathcal{P}$, then we will write

$$S := \{x : x \text{ satisfies } \mathcal{P}\}.$$

For instance,

$$\{x : x \text{ is prime}\} = \{2, 3, 5, 7, \dots\}$$

- Suppose S and S' are any two sets. The set $S \cup S'$ is the *union* of S and S' . It is the set

$$S \cup S' := \{x : x \in S \text{ or } x \in S'\}.$$

That is, $S \cup S'$ is the set of objects which are contained in at least 1 of S or S' . For instance,

$$\{x : x \text{ is divisible by } 3\} \cup \{x : x \text{ is even}\} = \{x : x \text{ is divisible by } 2 \text{ or } 3 \text{ (or both)}\}.$$

- Suppose S and S' are any two sets. The set $S \cap S'$ is the *intersection* of S and S' . It is the set

$$S \cap S' := \{x : x \in S \text{ and } x \in S'\}.$$

That is, $S \cap S'$ is the set of objects which are contained in both S and S' . For instance,

$$\{x : x \text{ is divisible by } 3\} \cap \{x : x \text{ is even}\} = \{x : x \text{ is divisible by } 6\}.$$

- If S is a set and S' is another set such that each $x \in S'$ also satisfies $x \in S$, then we say S' is a *subset* of S . We write this as $S' \subseteq S$. If S' is a subset of S and S is not a subset of S' , we write $S' \subset S$. We say that $S = S'$ if $S \subseteq S'$ and $S' \subseteq S$. For instance,

$$\{x : x \text{ is divisible by } 4\} \subseteq \{x : x \text{ is even}\}.$$

In fact, the containment is strict, i.e. replacing $\subseteq$ with $\subset$ in the statement above still results in a true statement.

- If S' is the subset of S such that each element of S' satisfies property $\mathcal{P}$, then we write

$$S' = \{x \in S : x \text{ satisfies } \mathcal{P}\}.$$

For instance, if S is the set of positive even numbers, then

$$\{x : x \text{ is divisible by } 6\} = \{x \in S : x \text{ is divisible by } 3\}.$$

- If S is a set and $x \in S$, the set $S \setminus \{x\}$ is the set

$$S \setminus \{x\} := \{y \in S : y \neq x\}.$$

That is, $S \setminus \{x\}$ is the subset of S which does not contain x . For example,

$$\{6, 7, 8\} \setminus \{8\} = \{6, 7\}.$$

- Generalizing the above, if S is a set and S' is any other set, then $S \setminus S'$ is the set

$$S \setminus S' := \{y \in S : y \notin S'\}.$$

That is, $S \setminus S'$ is the subset of S which does not contain any elements of S' . For instance, if S is the set of even numbers and

$$S' = \{x : x = 2n + 2 \text{ for some } n \in \mathbb{N}\}$$

then

$$S \setminus S' = \{x : x \text{ is divisible by } 4\}.$$

- If S is a set and S' is another set such that $S \subseteq S'$, we say that S' is a *superset* of S .
- The *cardinality* of a set S is the number of objects contained within the set. It is denoted as $|S|$.
- A *superset* of a set S is another set S' such that $S \subseteq S'$

Remark (Multisets): To be technically formal, when we are working with collections of objects containing possible duplicates, we define a *multiset* chosen from a set S as a map $\psi : S \rightarrow \mathbb{Z}_{\geq 0}$. For instance, if $S = \{a, b, c\}$ and $\psi(a) = 2$, $\psi(b) = 3$, and $\psi(c) = 0$, then this gives a multiset which we write as $\{a, a, b, b, b\}$. All of the set operations introduced previously also apply to multisets (so we can define submultisets, deletions from submultisets, etc.). However, you can treat them as if they were just normal sets, with duplicate elements.

- For sets A and B , $f : A \rightarrow B$ denotes a function from A to B , i.e. for any $a \in A$, the value $f(a)$ is an element of B .
- For $A' \subseteq A$, we also define $f(A') = \{f(a) : a \in A'\}$. For instance, if $f : \{0, 1, 3, 4\} \rightarrow \{6, 7\}$ is defined as $f(0) = f(1) = f(3) = 6$ and $f(4) = 7$, then $f(\{0, 1, 3\}) = \{6\}$.
- Similarly, for $b \in B$ we define $f^{-1}(b) = \{a \in A : f(a) = b\}$, and for $B' \subseteq B$ we define $f^{-1}(B') = \{a \in A : f(a) \in B'\}$. Taking the example above, we have $f^{-1}(6) = \{0, 1, 3\}$.
- $[k] := \{1, 2, \dots, k\}$ when k is a positive integer.
- We use

$$\max_{y \in S} f(y) \quad \text{and} \quad \min_{y \in S} g(y)$$

to indicate the maximum value of $f(y)$ and the minimum value of $g(y)$ among y in S , respectively. That is, if $S = \{1, 2, 3\}$ and $f : S \rightarrow \{4, 5, 6\}$ is defined as $f(1) = 4$, $f(2) = 6$ and $f(3) = 5$, then we would have

$$\max_{y \in S} f(y) = 6 \quad \text{and} \quad \min_{y \in S} f(y) = 4.$$

As a convention, from Sections 2 and on we will use $\mathcal{M}$ to denote matroids, X and Y to denote independent sets, P and Q to denote arbitrary subsets of E , and B to denote bases of $\mathcal{M}$. For ease of grading, it is recommended that you do the same.

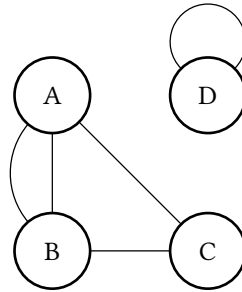
1. Introduction to Matroids

1.1. Graph Theory Basics

The central focus of this power round is on *matroids*. However, before we give any formal definitions related to matroids, it will be helpful to have a vivid picture in mind of what a matroid *looks like*. To this end, we will begin by looking at graphs. (We begin formally discussing matroids in Section 2; you may immediately skip there if you wish.)

Definition 1.1.1: A *graph* G is an ordered pair $G = (V(G), E(G))$ where $V(G)$ is a set containing elements called *vertices* and $E(G)$ is a multiset chosen from the set of unordered pairs of elements $V(G)$. (Elements can also be paired to themselves.) Each element in $E(G)$ is called an *edge*. If two edges connect the same two nodes, we call the edges *parallel*. When it is clear, we will write E and V for $E(G)$ and $V(G)$, respectively.

Here is an example of a graph. The set of nodes is $V = \{A, B, C, D\}$. The multiset of edges is $\{(A, B), (A, B), (B, C), (A, C), (D, D)\}$. Here, (A, B) and (A, B) are parallel edges.



Remark: We require the usage of multisets in the definition of graphs to account for possible duplicate edges. For all purposes, however, you can treat multisets as normal sets with duplicates.

Problem 1.1 [1 point].

Draw a graph with vertex set $V = \{A, B, C\}$ and edge set $E = \{(A, B), (A, C), (A, A)\}$.

An important notion in graph theory is a *cycle*. It can be thought of as a sequence of edges that one can walk along and arrive back at their starting position.

Definition 1.1.2:

- A (v_1, v_2) -*path* in a graph G is a sequence of distinct edges starting at $v_1 \in V$ and ending at $v_2 \in V$.
- A nonempty (v, v) -path is called a *cycle*.
- A subset $V' \subseteq V$ where there exists a path between any $v_1, v_2 \in V'$, but no path between any $v_1 \in V'$ and $v_2 \in V \setminus V'$, is called a *connected component*.
- We call G *connected* if V itself is a connected component.
- A graph having no cycles is called a *forest*, while a connected forest is called a *tree*.

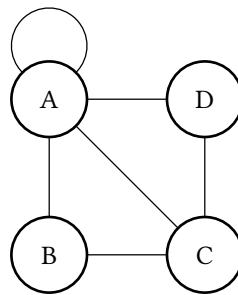
Problem 1.2 [2 points].

- (a) [1 point] Draw a graph with 4 vertices and 4 edges that is connected.
- (b) [1 point] Draw a graph with 4 vertices and 4 edges that has two connected components.

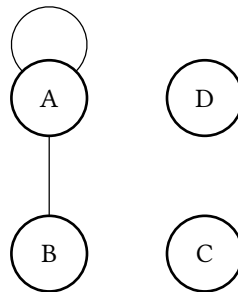
We can also think about *subgraphs* of a graph, which can be constructed by choosing to keep certain edges and discarding other ones.

Definition 1.1.3: Let $G = (V, E)$ be a graph. For a submultiset $E' \subseteq E$ we define the *induced subgraph* $G[E'] = (V, E')$. In particular, if $G[E']$ contains a cycle we say E' *induces a cycle*, and otherwise we say E' *induces a forest*.

Example: Consider the following graph G with vertex set $\{A, B, C, D\}$ and edge multiset $\{(A, A), (A, B), (A, C), (A, D), (B, C), (C, D)\}$.



If we take the submultiset $E' = \{(A, A), (A, B)\}$, then the induced subgraph $G[E']$ is the following graph.

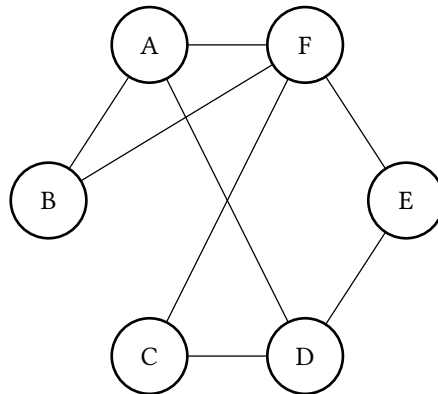


Problem 1.3 [3 points].

- (a) [1 point] In the example above, does E' induce a forest? (You do not need to provide proof.)
- (b) [1 point] Another possible submultiset of E in the example above is $E'' = \{(A, B), (B, C)\}$. Draw the induced subgraph $G[E'']$.
- (c) [1 point] Does E'' induce a forest? (You do not need to provide proof.)

Problem 1.4 [3 points].

Consider the following graph G .



- (a) [1 point] Identify a nonempty submultiset E' of $E(G)$ which induces a cycle.
- (b) [2 points] Identify a submultiset E' of $E(G)$ which induces a forest with the greatest number of edges. That is, find a submultiset E' which induces a forest such that for any other submultiset $E'' \subseteq E$ which induces a forest, we have $|E''| \leq |E'|$.

Problem 1.5 [2 points].

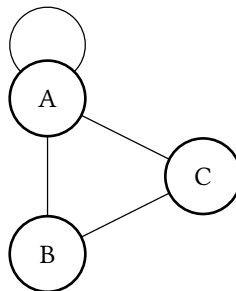
Let $G = (V, E)$ be a graph, and suppose $E' \subseteq E$ induces a forest. Show that E' contains no parallel edges (so E' is a genuine set).

1.2. Hereditary and Exchange Properties

Given a graph $G = (V, E)$, we are interested in the submultisets which induce forests. By Problem 1.5, these submultisets are actually sets. They also satisfy two important properties: the *hereditary* and *exchange* properties.

Problem 1.6 [2 points].

Consider the following graph $G = (V, E)$.



Write down all of the subsets of E which induce forests. (Include the empty set.)

Problem 1.7 [3 points].

Let $G = (V, E)$ be a graph, and suppose $F \subseteq E$ induces a forest. If $F' \subseteq F$, show that F' also induces a forest.

We can think of all of the subsets you identified in Problem 1.6 (i.e. all of the subsets of E which induce forests) as a *set of sets*. Problem 1.7 then shows that if F is contained in this set of sets and $F' \subseteq F$, then F' is also contained in the set of sets. We make this formal in the following definition.

Definition 1.2.1: Let $\mathcal{I}$ be a set of sets over some ground set E . (That is, each set $X \in \mathcal{I}$ is a subset of E .) Suppose that if $Y \in \mathcal{I}$ and $X \subseteq Y$, then $X \in \mathcal{I}$ as well. Then, we say that $\mathcal{I}$ satisfies the *hereditary property*.

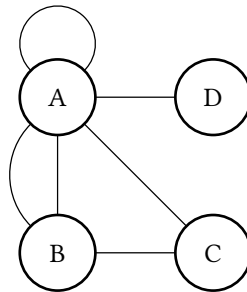
In other words, we have shown that for a graph $G = (V, E)$, the set

$$\mathcal{I} := \{E' \subseteq E : E' \text{ induces a forest}\}$$

satisfies the hereditary property. The set $\mathcal{I}$ also satisfies a different property, as we will now discuss.

Problem 1.8 [3 points].

Consider the following graph $G = (V, E)$.



- (a) [1 point] Find a subset F of E such that $|F| = 3$ and F induces a forest.
- (b) [1 point] Find a subset F' of E such that $F' \not\subseteq F$, $|F'| < |F|$, and F' induces a forest.
- (c) [1 point] Find an edge $e \in F \setminus F'$ such that the set $F' \cup \{e\}$ induces a forest.

This turns out to be another general property of the set of subsets of E which induce forests.

Definition 1.2.2: Let $\mathcal{I}$ be a set of sets over some ground set E . Suppose that if $X, Y \in \mathcal{I}$ and $|X| < |Y|$, then there exists $e \in Y \setminus X$ such that $X \cup \{e\} \in \mathcal{I}$ as well. Then, we say that $\mathcal{I}$ satisfies the *exchange property*.

Problem 1.9 [2 points].

Consider $\mathcal{I} = \{\emptyset, \{1\}, \{2\}, \{1, 2, 3\}\}$. Does $\mathcal{I}$ satisfy the exchange property? If so, write “Yes.” If not, write down the smallest collection of set(s) that need to be added to $\mathcal{I}$ in order to make it satisfy the exchange property.

We will now show that the set of subsets of E which induce forests (which we will denote as $\mathcal{I}$) actually satisfies the exchange property. The following fact will be useful.

Proposition 1.2.1: If $G = (V, E)$ is a forest, then it has $|V| - |E|$ connected components.

Problem 1.10 ($\mathcal{I}$ satisfies exchange property) [6 points].

- (a) [3 points] Suppose $F, F' \in \mathcal{I}$ with $|F| < |F'|$. Show that $G[F \cup F']$ has strictly fewer connected components than $G[F]$.
- (b) [3 points] Deduce that $\mathcal{I}$ satisfies the exchange property, i.e. that there exists $e \in F' \setminus F$ such that $F \cup \{e\} \in \mathcal{I}$.

The properties which we have outlined here turn out to be extremely desirable, which motivates the following definition.

Definition 1.2.3: If E is any finite set, a *matroid* $\mathcal{M}$ is a pair $\mathcal{M} = (E, \mathcal{I})$, where $\mathcal{I}$ is a set of sets over E such that:

- (a) $\emptyset \in \mathcal{I}$.
- (b) $\mathcal{I}$ satisfies the hereditary property. That is, if $X \in \mathcal{I}$ and $X' \subseteq X$, then $X' \in \mathcal{I}$.
- (c) $\mathcal{I}$ satisfies the exchange property. That is, if $X_1, X_2 \in \mathcal{I}$ and $|X_1| < |X_2|$, there exists $e \in X_2 \setminus X_1$ such that $X_1 \cup \{e\} \in \mathcal{I}$.

The sets $X \in \mathcal{I}$ are called *independent*, while all $X' \subseteq E$ such that $X' \notin \mathcal{I}$ are called *dependent*.

(For clarity, we will sometimes write $\mathcal{I}(\mathcal{M})$ and $E(\mathcal{M})$ for $\mathcal{I}$ and E if more than one matroid is present.)

The preceding discussion shows the following theorem.

Theorem 1.2.2: If $G = (V, E)$ is a graph and

$$\mathcal{I} = \{X \subseteq E : X \text{ induces a forest}\}$$

then the pair $(E, \mathcal{I})$ forms a matroid (called the *graphic matroid*).

In this language, the sets of edges of a graph G which induce forests are exactly the independent sets of the graphic matroid.

2. Examples and Properties of Matroids

For convenience, we repeat the definition of a matroid here.

Definition 2.1: If E is any finite set, a *matroid* $\mathcal{M}$ is a pair $\mathcal{M} = (E, \mathcal{I})$, where $\mathcal{I}$ is a set of sets over E such that:

- (a) $\emptyset \in \mathcal{I}$.
- (b) $\mathcal{I}$ satisfies the hereditary property. That is, if $X \in \mathcal{I}$ and $X' \subseteq X$, then $X' \in \mathcal{I}$.
- (c) $\mathcal{I}$ satisfies the exchange property. That is, if $X_1, X_2 \in \mathcal{I}$ and $|X_1| < |X_2|$, there exists $e \in X_2 \setminus X_1$ such that $X_1 \cup \{e\} \in \mathcal{I}$.

The sets $X \in \mathcal{I}$ are called *independent*, while all $X' \subseteq E$ such that $X' \notin \mathcal{I}$ are called *dependent*.

(For clarity, we will sometimes write $\mathcal{I}(\mathcal{M})$ and $E(\mathcal{M})$ for $\mathcal{I}$ and E if more than one matroid is present.)

Theorem 1.2.2 gives us one important example of a matroid. In general, to check that a given pair $(E, \mathcal{I})$ is a matroid, we need to check that $\mathcal{I}$ contains the empty set and that $\mathcal{I}$ satisfies the hereditary and exchange properties.

Example: Let $E = \{1, 2, 3\}$ and $\mathcal{I} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. We verify that $(E, \mathcal{I})$ is a matroid. Indeed, $\emptyset \in \mathcal{I}$ is given. To verify that $\mathcal{I}$ satisfies the hereditary property, we note that all possible subsets of the set $\{1, 2\}$ are

$$\emptyset, \{1\}, \{2\}, \{1, 2\}$$

which are all contained in $\mathcal{I}$. Similarly, all possible subsets of the set $\{1\}$ are $\emptyset, \{1\}$, which are all contained in $\mathcal{I}$, and all possible subsets of the set $\{2\}$ are $\emptyset, \{2\}$, which are all contained in $\mathcal{I}$.

To verify the exchange property, we note that if two sets in $\mathcal{I}$ have differing cardinalities, then the cardinalities of the sets must either be 0 and 1, 1 and 2, or 0 and 2. In the latter case, the only set with cardinality 2 is $\{1, 2\}$ and the only set with cardinality 0 is $\emptyset$. Adding 1 or 2 to $\emptyset$ gives us another element of $\mathcal{I}$, so this takes care of the last case. In the first case, the sets of cardinality 1 are $\{1\}$ and $\{2\}$, and adding 1 or 2 to $\emptyset$ gets another element of $\mathcal{I}$. Finally, in the second case, the set of cardinality 2 is $\{1, 2\}$ and the sets of cardinality 1 are $\{1\}$ and $\{2\}$. Adding 2 to $\{1\}$ gives the set $\{1, 2\}$ which is contained in $\mathcal{I}$, and adding 1 to $\{2\}$ gives the set $\{1, 2\}$ again. This takes care of all of the cases, and we are done. (Usually the process is not so tedious; we are only being exhaustive for clarity.)

Problem 2.1 [2 points].

Consider the set $E = \{1, 2, 3, 4\}$ and $\mathcal{I} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, \{2, 4\}, \{1, 2, 4\}\}$. Is $\mathcal{M} = (E, \mathcal{I})$ a matroid? If so, answer “Yes.” If not, write down which set(s) need to be added to $\mathcal{I}$ to make $\mathcal{M}$ a matroid.

Let us now construct other examples of matroids.

Problem 2.2 (Uniform matroid) [3 points].

Let $E = \{1, 2, \dots, 10\}$. Fix a nonnegative integer $n \leq 10$, and suppose $\mathcal{I}$ is the set of $X \subseteq E$ such that $|X| \leq n$. Verify that $\mathcal{M} = (E, \mathcal{I})$ is a matroid.

We can also construct new matroids from matroids we already have.

Definition 2.2: Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid. Pick a subset P of the elements E . We define

$$\mathcal{I}|P := \{X \subseteq P : X \in \mathcal{I}\}.$$

Then the pair $\mathcal{M}|P := (P, \mathcal{I}|P)$ is called the *restriction* of $\mathcal{M}$ to P .

Problem 2.3 (*Restriction of a matroid*) [3 points].

Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid and $P \subseteq E$. Prove that the restriction $\mathcal{M}|P$ is a matroid.

In addition to Theorem 1.2.2, here is another example of a matroid from a graph.

Definition 2.3: Let E be a ground set and $A_1, \dots, A_k \subseteq E$ be sets, not necessarily disjoint, for some positive integer k . We will call $X \subseteq E$ independent if there exists $\varphi : X \rightarrow [k]$ such that $|X| = |\varphi(X)|$, and $x \in A_{\varphi(x)}$ for every $x \in X$. If $\mathcal{I}$ is the set of all independent sets, we call the pair $\mathcal{M} = (E, \mathcal{I})$ a *transversal matroid*.

You will show later on that this actually defines a matroid; however, here we simply present it as a result that you may use. (You may need it for future sections.)

Theorem 2.1: A transversal matroid is a matroid.

2.1. Bases of a Matroid

Matroids are characterized by their independent sets, but it turns out we can also characterize matroids using their *maximal* independent sets. To make this precise, we make the following definition.

Definition 2.1.1: A *basis* of a matroid $\mathcal{M} = (E, \mathcal{I})$ is a maximal independent set $B \in \mathcal{I}$. That is, B is a basis if any strict superset of B is not independent. We denote the collection of all bases of $\mathcal{M}$ as $\mathcal{B}(\mathcal{M})$ (or just $\mathcal{B}$ when $\mathcal{M}$ is unambiguous).

The following problem shows that starting with $\mathcal{B}$ and going “down” (removing elements from bases), we can generate all independent sets.

Problem 2.4 [4 points].

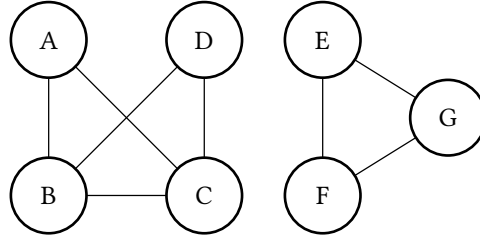
Show that for any $X \in \mathcal{I}$, there exists a basis $B \in \mathcal{B}$ such that $X \subseteq B$.

Problem 2.5 [2 points].

Show that if $B_1, B_2 \in \mathcal{B}$ then $|B_1| = |B_2|$.

Problem 2.6 [4 points].

Consider the following graph G .



- (a) [2 points] Let $\mathcal{M}$ be the graphic matroid of G . Find a basis of $\mathcal{M}$. Justify your answer.
 (b) [2 points] Compute the number of bases of $\mathcal{M}$.

In Section 5 we will discuss algorithms to compute such bases. In particular, if each element $e \in E$ is given a weight w_e , it can be useful to find the max- or min-weight basis—for example, in a graphic matroid, this corresponds to a minimum spanning tree.

Problem 2.7 [4 points].

Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid with $E = \{1, 2, \dots, 10\}$. Find the maximum number of bases of $\mathcal{M}$.

2.2. Rank of a Matroid

Problem 2.5 allows us to make the following definition unambiguously, which gives us algebraic quantities we can work with.

Definition 2.2.1: Let 2^E denote the *power set* of E , which is the set of all possible subsets of E . For a matroid $\mathcal{M} = (E, \mathcal{I})$, define a function $r_{\mathcal{M}} : 2^E \rightarrow \mathbb{Z}_{\geq 0}$ such that for any $P \in 2^E$, $r_{\mathcal{M}}(P)$ returns the cardinality of a basis of $\mathcal{M}|P$. (Recall $\mathcal{M}|P$ is the restriction of $\mathcal{M}$ to P .) We call $r_{\mathcal{M}}$ the *rank function* of $\mathcal{M}$.

It's clear that if $P \subseteq E$, then $0 \leq r_{\mathcal{M}}(P) \leq |P|$. The rank function also satisfies the following properties.

Problem 2.8 (*Rank is increasing*) [3 points].

If $P \subseteq Q \subseteq E$, show that $r_{\mathcal{M}}(P) \leq r_{\mathcal{M}}(Q)$.

Problem 2.9 [6 points].

If $P \subseteq Q \subseteq E$ and $e \in E$, show that

$$r_{\mathcal{M}}(Q \cup \{e\}) - r_{\mathcal{M}}(Q) \leq r_{\mathcal{M}}(P \cup \{e\}) - r_{\mathcal{M}}(P).$$

Problem 2.10 (*Rank is submodular*) [6 points].

Let matroid $\mathcal{M} = (E, \mathcal{I})$ have rank function $r_{\mathcal{M}}$. Show that $r_{\mathcal{M}}$ is *submodular*, i.e.

$$r_{\mathcal{M}}(P) + r_{\mathcal{M}}(Q) \geq r_{\mathcal{M}}(P \cup Q) + r_{\mathcal{M}}(P \cap Q)$$

for any $P, Q \subseteq E$.

We now prove Theorem 2.1.

Problem 2.11 (*Transversal matroid is a matroid*) [10 points].

Show that Definition 2.3 defines a matroid.

3. Matroid Intersection

A natural question to ask about matroids is: how can we combine them and still get another matroid? Here, we are interested in the intersection of two matroids over the same ground set E .

Definition 3.1: Let $\mathcal{M}_1 = (E, \mathcal{I}_1)$ and $\mathcal{M}_2 = (E, \mathcal{I}_2)$ be matroids. The *matroid intersection* of $\mathcal{M}_1$ and $\mathcal{M}_2$, denoted $\mathcal{M}_1 \cap \mathcal{M}_2$, is the pair

$$\mathcal{M}_1 \cap \mathcal{M}_2 = (E, \mathcal{I}_1 \cap \mathcal{I}_2).$$

Intersections of matroids allow us to do some powerful things.

Problem 3.1 (*Bipartite matching*) [2 points].

Let $G = (V, E)$ be a graph. Suppose there exist disjoint V_1 and V_2 such that $V = V_1 \cup V_2$, and all edges in E are of the form (v_1, v_2) for $v_1 \in V_1$ and $v_2 \in V_2$. (In particular there are no edges within V_1 and no edges within V_2 .)

Let $\mathcal{M}_1 = (E, \mathcal{I}_1)$ be a matroid where we say $X \subseteq E$ is independent if every $v_1 \in V_1$ is an endpoint of at most one edge in X . Verify that $\mathcal{M}_1$ is a matroid.

In the context of Problem 3.1, if we define $\mathcal{M}_2$ similarly for V_2 , then the maximum-size common independent set of $\mathcal{M}_1$ and $\mathcal{M}_2$ is called a maximum matching on G . This has real-life usage: for example, suppose V_1 is the set of power round writers, V_2 is the set of toothpaste flavors, and E is a set of preferences (v_1, v_2) where person v_1 likes flavor v_2 . Then if only one tube of each toothpaste flavor is available, a maximum matching would be the largest number of power round writers who could receive a flavor of toothpaste they like.

On the other hand, the intersection of two matroids need not be a matroid, as shown by the following problem.

Problem 3.2 [2 points].

Let $E = \{1, 2, 3\}$. Let $\mathcal{I}_1 = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}\}$ and $\mathcal{I}_2 = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 3\}, \{2, 3\}\}$.

- (a) [1 point] Verify that $\mathcal{M}_1 = (E, \mathcal{I}_1)$ and $\mathcal{M}_2 = (E, \mathcal{I}_2)$ are matroids.
- (b) [1 point] Verify that $\mathcal{M}_1 \cap \mathcal{M}_2$ is not a matroid.

Here is a very challenging problem. You will not need it for any other problem, so we recommend only attempting it if you have extra time.

Problem 3.3 [28 points].

Let $E = \{1, 2, \dots, 10\}$. Say a matroid $\mathcal{M}$ on E is *strong* if $\mathcal{M} \cap \mathcal{M}'$ is a matroid for every matroid $\mathcal{M}'$ on E .

- (a) [8 points] Show there exist 5121 strong matroids $\mathcal{M}$. (Partial credit will be given for finding 1024 such matroids.)
- (b) [20 points] Show all other matroids $\mathcal{M}$ are not strong.

Nevertheless, there are many useful things we can do with matroid intersections. For example, finding the maximum-size bipartite matching is a useful general problem, and it can be phrased as finding the maximum-size common basis of $\mathcal{M}_1$ and $\mathcal{M}_2$ as in Problem 3.1.

The following problem gives a relatively easy upper bound on this maximum-size common basis.

Problem 3.4 [6 points].

Show that

$$\max_{X \in \mathcal{I}_1 \cap \mathcal{I}_2} |X| \leq \min_{P \subseteq E} [r_{\mathcal{M}_1}(P) + r_{\mathcal{M}_2}(E \setminus P)].$$

The incredible thing is that the inequality in Problem 3.4 is actually an equality! In practice, this fact is used to solve matroid intersection problems. We will prove this equality in the next several problems. The following way of constructing new matroids will be important for us.

Definition 3.2: Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid and $e \in E$. The *deletion* of e from $\mathcal{M}$, denoted $\mathcal{M} \setminus e$, is the matroid $\mathcal{M}|(E \setminus \{e\})$. For notational simplicity, we denote $\mathcal{I} \setminus e = \mathcal{I}|(E \setminus \{e\})$, so $\mathcal{M} \setminus e = (E \setminus \{e\}, \mathcal{I} \setminus e)$.

If $r_{\mathcal{M}}(\{e\}) = 1$, we define $\mathcal{I}/e := \{X \subseteq E \setminus \{e\} : X \cup \{e\} \in \mathcal{I}\}$. The *contraction* of $\mathcal{M}$ with e , denoted as $\mathcal{M}/e$, is defined as the pair

$$\mathcal{M}/e := (E \setminus \{e\}, \mathcal{I}/e).$$

Problem 3.5 (Contraction matroid is a matroid) [4 points].

Suppose $\mathcal{M} = (E, \mathcal{I})$ is a matroid and $e \in E$ has $r_{\mathcal{M}}(\{e\}) = 1$. Show that $\mathcal{M}/e$ is a matroid.

The following problem shows how the rank function interacts with the contraction operation.

Problem 3.6 (Rank function after contraction) [4 points].

Suppose $\mathcal{M} = (E, \mathcal{I})$ is a matroid and $e \in E$ has $r_{\mathcal{M}}(\{e\}) = 1$. For any $P \subseteq E \setminus \{e\}$, show that

$$r_{\mathcal{M}/e}(P) = r_{\mathcal{M}}(P \cup \{e\}) - 1.$$

We will now use these tools to prove the matroid intersection theorem.

Problem 3.7 [17 points].

In this problem, we will prove that there exists a common independent set $X \in \mathcal{I}_1 \cap \mathcal{I}_2$ of size

$$k := \min_{P \subseteq E} [r_{\mathcal{M}_1}(P) + r_{\mathcal{M}_2}(E \setminus P)].$$

We proceed by induction on the number z of elements $e \in E$ such that $r_{\mathcal{M}_1}(\{e\}) = r_{\mathcal{M}_2}(\{e\}) = 1$.

- (a) [4 points] Show that the statement holds if $z = 0$.
- (b) [8 points] Now suppose there is $e \in E$ such that $r_{\mathcal{M}_1}(\{e\}) = r_{\mathcal{M}_2}(\{e\}) = 1$, and for contradiction suppose that there exists no common independent set in $\mathcal{I}_1 \cap \mathcal{I}_2$ of size k . Show that there exist sets $P, Q \subseteq E \setminus \{e\}$ such that

$$r_{\mathcal{M}_1}(P) + r_{\mathcal{M}_2}((E \setminus \{e\}) \setminus P) \leq k - 1$$

and

$$r_{\mathcal{M}_1}(Q \cup \{e\}) + r_{\mathcal{M}_2}(E \setminus Q) \leq k.$$

(Hint: consider applying the induction hypothesis to the intersections $(\mathcal{M}_1 \setminus e) \cap (\mathcal{M}_2 \setminus e)$ and $(\mathcal{M}_1/e) \cap (\mathcal{M}_2/e)$.)

- (c) [5 points] Using the inequalities in (b), complete the induction on z .

Combining Problem 3.7 and Problem 3.4, we have proved the following theorem.

Theorem 3.1 (Matroid intersection theorem): If $\mathcal{M}_1 = (E, \mathcal{I}_1)$ and $\mathcal{M}_2 = (E, \mathcal{I}_2)$ are matroids defined on the same ground set E , then

$$\max_{X \in \mathcal{I}_1 \cap \mathcal{I}_2} |X| = \min_{P \subseteq E} [r_{\mathcal{M}_1}(P) + r_{\mathcal{M}_2}(E \setminus P)].$$

Remark (*Computability theory, not needed for rest of power round*): Finding the size of a basis of a matroid is a common type of problem, and is solvable in polynomial time. Theorem 3.1 can be modified to find the maximum size of a common basis of two matroids in polynomial time. On the other hand, three-matroid intersection is already NP-hard, and there is no (known) corresponding statement like Theorem 3.1 for this case.

4. Matroid Union

Let us now look at taking the unions of matroids. Unlike intersections, this actually gives us a matroid. This has a number of surprising consequences, as we will see.

Problem 4.1 [6 points].

Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid and E' be any set. For any map $f : E \rightarrow E'$, let $\mathcal{M}' = (E', \mathcal{I}')$, where

$$\mathcal{I}' = \{f(X) : X \in \mathcal{I}\}.$$

- (a) [1 point] Show that for any $X' \in \mathcal{I}'$, there exists $X \in \mathcal{I}$ such that $f(X) = X'$ and $|X| = |X'|$.
- (b) [1 point] Show that $\emptyset \in \mathcal{I}'$, and verify that $\mathcal{M}'$ satisfies the hereditary property.
- (c) [4 points] Show that $\mathcal{M}'$ satisfies the exchange property.

It follows that $\mathcal{M}'$ is a matroid.

Problem 4.1 interacts particularly nicely with the following definition.

Definition 4.1: Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid and E' be any set, and let $f : E \rightarrow E'$ be a map. Fix $P' \subseteq E'$ and let $P = f^{-1}(P')$. Define

$$\mathcal{I}_p(P) := \{Q \subseteq P : |Q| = |f(Q)|\}.$$

The pair $\mathcal{M}_p(P) := (P, \mathcal{I}_p(P))$ is called the *partition matroid* induced by $P = f^{-1}(P')$. When it is clear, we will just write $\mathcal{M}_p$ and $\mathcal{I}_p$ for $\mathcal{M}_p(P)$ and $\mathcal{I}_p(P)$.

Problem 4.2 (*Partition matroid is a matroid*) [5 points].

Inherit notation from the previous definition.

- (a) [2 points] Verify that $\mathcal{M}_p(P)$ is a matroid. (*Hint:* You may use Problem 2.11.)
- (b) [3 points] Show that for any $Q \subseteq P$, we have

$$r_{\mathcal{M}_p(P)}(Q) = |\{e' \in P' : f^{-1}(e') \cap Q \neq \emptyset\}|.$$

Problem 4.3 [10 points].

- (a) [4 points] Let $\mathcal{M} = (E, \mathcal{I})$ and $f : E' \rightarrow E$ be a map, and let $\mathcal{M}' = (E', \mathcal{I}')$ be the matroid constructed in Problem 4.1. Let $P' \subseteq E'$, and denote $P = f^{-1}(P')$. Also, let $\mathcal{M}_p = (P, \mathcal{I}_p)$ be the partition matroid induced by P . Show that the rank function for $\mathcal{M}'$ satisfies

$$r_{\mathcal{M}'}(P') = \max_{X \in \mathcal{I}' \cap P'} |X|.$$

- (b) [6 points] Show that

$$r_{\mathcal{M}'}(P') = \min_{Q' \subseteq P'} [r_{\mathcal{M}}(f^{-1}(Q')) + |P' \setminus Q'|].$$

The following definition gives us a natural way of how to take the union of matroids.

Definition 4.2: Let $\mathcal{M}_1 = (E_1, \mathcal{I}_1), \dots, \mathcal{M}_k = (E_k, \mathcal{I}_k)$ be matroids. Let $E = \bigcup_{i=1}^k E_i$ and

$$\mathcal{I} = \left\{ \bigcup_{i=1}^k X_i : X_1 \in \mathcal{I}_1, \dots, X_k \in \mathcal{I}_k \right\}.$$

The *matroid union* of $\mathcal{M}_1, \dots, \mathcal{M}_k$, denoted as $\mathcal{M}_1 \vee \mathcal{M}_2 \vee \dots \vee \mathcal{M}_k$, is defined as

$$\mathcal{M}_1 \vee \mathcal{M}_2 \vee \dots \vee \mathcal{M}_k := (E, \mathcal{I}).$$

Problem 4.4 (*Matroid union is a matroid*) [7 points].

Suppose $\mathcal{M}_1 = (E_1, \mathcal{I}_1), \dots, \mathcal{M}_k = (E_k, \mathcal{I}_k)$ are matroids, and let $\mathcal{M} = \mathcal{M}_1 \vee \dots \vee \mathcal{M}_k$.

- (a) [3 points] Given $E_i \cap E_j = \emptyset$ for all $i \neq j$, show that $\mathcal{M}$ is a matroid.
- (b) [4 points] Prove (a) without the $E_i \cap E_j = \emptyset$ condition.

Problem 4.5 (*Rank function of matroid union*) [4 points].

If $\mathcal{M}_1 = (E_1, \mathcal{I}_1), \dots, \mathcal{M}_k = (E_k, \mathcal{I}_k)$ are matroids and $\mathcal{M} = \mathcal{M}_1 \vee \mathcal{M}_2 \vee \dots \vee \mathcal{M}_k$, show that for any $P \subseteq \bigcup_{i=1}^k E_i$ we have

$$r_{\mathcal{M}}(P) = \min_{Q \subseteq P} \left[\sum_{i=1}^k r_{\mathcal{M}_i}(Q \cap E_i) + |P \setminus Q| \right].$$

Here are several applications of matroid unions. The original proofs of these propositions did not make use of matroid unions and were quite difficult.

Problem 4.6 (*Edmonds, Nash-Williams*) [16 points].

- (a) [4 points] Let k be an arbitrary positive integer. Show that a matroid $\mathcal{M} = (E, \mathcal{I})$ has k disjoint bases if and only if, for every subset $P \subseteq E$, we have

$$kr_{\mathcal{M}}(P) + |E \setminus P| \geq kr_{\mathcal{M}}(E).$$

- (b) [6 points] Show that a matroid $\mathcal{M}$ has k independent sets whose union is E if and only if, for every subset $P \subseteq E$, we have

$$kr_{\mathcal{M}}(P) \geq |P|.$$

- (c) [6 points] Let $G = (V(G), E(G))$ be a graph. For $P \subseteq V(G)$, let $G|P$ denote the graph with vertex set P and edge set

$$E|P := \{e \in E : e \text{ is incident to a vertex } v \in P\}.$$

Show that G admits k forests whose union is $E(G)$ if and only if, for all subsets $P \subseteq V(G)$, we have

$$|E(G|P)| \leq k(|P| - 1).$$

(We say that an edge $e = (v_1, v_2)$ is *incident* to a vertex w if $v_1 = w$ or $v_2 = w$.)

5. Greedy Algorithms

Here is an application of matroids in the context of finding a maximum-weight basis. Finding such a basis turns out to be very useful not only in mathematics, but also in asset allocation and even bioinformatics.

5.1. The Greedy Algorithm for Matroids

Definition 5.1.1: An *algorithm* is a step-by-step set of instructions followed in order to accomplish a specific, predefined task.

For example, an algorithm with the objective of brushing one's teeth might involve getting gas, driving to a department store, buying toothpaste and a toothbrush, opening the toothpaste, adding toothpaste to the end of the toothbrush, lifting the toothbrush to your mouth, etc. Of course, this depends on the information and resources one has when executing the task. If you already have toothpaste, for instance, it doesn't make much sense to drive to a department store to buy more.

When writing algorithms, we will adopt the following terminology.

- **for:** Executes tasks that are in between the **for** and the next consecutive **end for** statement, until the conditions of the **for** loop are met. For instance, **for** $i \leftarrow 1$ to n will repeat whatever is contained within the **for** block n times, once for each $i \in \{1, 2, \dots, n\}$ in this order.
- **if:** Executes tasks that are in between the **if** and the next consecutive **end if**, if the conditions of the **if** statement are met.
- $a \leftarrow b$: Assigns whatever data is stored at b and copies it onto a .

In the following algorithm, given a matroid $\mathcal{M} = (E, \mathcal{I})$ we will assign a *weight* $w_e \in \mathbb{R}$ to each element $e \in E$. Then, for $P \subseteq E$ we will set $w(P) := \sum_{e \in P} w_e$.

Algorithm 5.1.1: Consider the following algorithm.

Input: A matroid $\mathcal{M} = (E, \mathcal{I})$, where $E = \{1, 2, \dots, n\}$, and the weight of each $e \in E$ is w_e .

Output: A basis $\hat{B} \in \mathcal{I}$ such that $w(\hat{B}) = \max_{B \in \mathcal{B}} w(B)$

1. Relabel the elements of the matroid so that $w_1 \geq w_2 \geq \dots \geq w_n$
2. $S \leftarrow \emptyset$
3. **for** $i \leftarrow 1$ to n **do**:
4. **if** $S \cup \{i\} \in \mathcal{I}$ **then**
5. $S \leftarrow S \cup \{i\}$
6. **end if**
7. **end for**
8. **return** S

We will call this algorithm the *greedy algorithm*.

It is surprising that this algorithm can even find a maximum-weight basis! This turns out to be equivalent to the fact that the underlying structure is that of a matroid, as the next theorem shows.

Theorem 5.1.1 (Rado, Gale): For any ground set $E = \{1, 2, \dots, n\}$, and a family of subsets $\mathcal{I}$ over E satisfying the hereditary property, the greedy algorithm returns a maximum-weight independent set for any set of positive weights $w : E \rightarrow \mathbb{R}^+$ if and only if $\mathcal{M} = (E, \mathcal{I})$ is a matroid.

We will now prove Theorem 5.1.1.

Problem 5.1 [8 points].

Prove the reverse direction of Theorem 5.1.1. That is, prove that if $(E, \mathcal{I})$ is a matroid, then for any set of positive weights assigned to the elements of E , the greedy algorithm returns a maximum-weight independent set.

Problem 5.2 [12 points].

Prove the forward direction of Theorem 5.1.1. That is, prove that if $(E, \mathcal{I})$ does not satisfy the exchange property, then there exists an assignment of positive weights to the elements of E such that the greedy algorithm does not return a maximum-weight independent set.